Asset Pricing Theory: S03 Dynamic Consumption and Portfolio Choice

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Structure of the course

- Email: amedeo.andriollo@dauphine.psl.eu
- Grading follows a 30-70 rule: 70% final exam, 30% project/homework (7th sessions).
- Office hours: feel "free" to (DO) come: B612.
- Slides/materials are updated regularly. Report a typo/mistake: corrections help the class.
- This session:
  - Continuous-time returns, Itô's lemma, dynamic programming, the HJB, and the Merton consumption–portfolio problem.
  - Main references: Cochrane (2005, Sec. 1.5 and App. A); Campbell (2018, Chs. 2 and 9); Merton (1971).

▷ Next session: Production-Based Asset Pricing.

Today's Route: Incorporating Dynamics

- Session 2 linked asset prices to marginal utility: $M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}$, and $1 = \mathbb{E}_t[M_{t+1} R^g_{i,t+1}]$.
- This session makes consumption and portfolio choice **jointly endogenous over** continuous **time**.

Discrete time: finite-period returns

$$R^g_{t,t+1} = \frac{S_{t+1} + D_{t+1}}{S_t}.$$

- Payoffs and rebalancing occur over nonzero intervals.
- Gross returns compound over the interval.

Continuous time: local evolution

$$\frac{\mathrm{d}S_t}{S_t} = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}B_t.$$

- Decisions and shocks arrive continuously.
- Drift is order $\mathrm{d}t$; risk is order $\sqrt{\mathrm{d}t}$.
- $\mathrm{d}S_t/S_t$ is an instantaneous simple return, not a discrete gross return.



The Investment Opportunity Set

- The investor can allocate wealth between a locally risk-free account A_t and a risky asset S_t.
- The money-market account earns the safe rate r (with its return is known at date t):

$$\frac{\mathrm{d}A_t}{A_t} = r\,\mathrm{d}t.$$

- The risky asset follows

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}B_t,$$

  i) μ is the expected instant. return, ii) $\sigma > 0$ is volatility, and iii) $\mathrm{d}B_t$ is the Brownian innovation.
- Over a short interval Δt, this implies approximately

$$\frac{\Delta S_t}{S_t} \approx \mu\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_{t+\Delta t}, \qquad \varepsilon_{t+\Delta t} \sim N(0,1).$$

  Thus expected return accumulates at rate Δt, while uncertainty accumulates at rate $\sqrt{\Delta t}$.
- The compensation for bearing (local) risk is summarized by $\frac{\mu - r}{\sigma}$, i.e., the instantaneous Sharpe ratio (expected excess return per unit of volatility).

Diffusion Model

- A diffusion model has the form: $\mathrm{d}X_t = a(X_t, u_t, t)\,\mathrm{d}t + b(X_t, u_t, t)\,\mathrm{d}z_t$,
  where X_t is the state variable and u_t is the control chosen using information available at date t.
- The drift term determines the mean, while the diffusion term b determines the scale (Brownian shock):

$$\mathbb{E}_t[\mathrm{d}X_t] = a\,\mathrm{d}t, \qquad \mathrm{Var}_t(\mathrm{d}X_t) = b^2\,\mathrm{d}t.$$

- Brownian increments over a finite interval: $z_{t+\Delta} - z_t \sim N(0, \Delta)$.
  $\hookrightarrow$ we write

$$z_{t+\Delta} - z_t = \sqrt{\Delta} \cdot \varepsilon_{t+\Delta}, \qquad \varepsilon_{t+\Delta} \sim N(0,1).$$

- Over a short interval,

$$\Delta X_t \approx a\,\Delta + b\sqrt{\Delta}\,\varepsilon_{t+\Delta}.$$

  hence the predictable component is order Δ, while the random component is order $\sqrt{\Delta}$.
- As $\Delta \downarrow 0$, $\sqrt{\Delta} \gg \Delta$. Hence, over a very short interval, the (realized) changes in X_t are typically driven more by the random mean-zero shock than by the drift.
  $\hookrightarrow$ Locally, uncertainty dominates what we **observe**, while the drift determines what we **expect**.

Why Does a Second-Order Term Survive?

- For a smooth function $f(X_t)$, a Taylor expansion over a short interval: $\Delta f \approx f_X \Delta X + \frac{1}{2} f_{XX} (\Delta X)^2$.
- Suppose X follows the diffusion model: $\Delta X = a\Delta t + b\Delta z$, $\Delta z \sim N(0, \Delta t)$.
  Since $\Delta z = \sqrt{\Delta t}\,\varepsilon$ with $\varepsilon \sim N(0,1)$,

$$\Delta z = O(\sqrt{\Delta t}), \qquad (\Delta z)^2 = O(\Delta t).$$

- Let us look at the quadratic term,

$$(\Delta X)^2 = a^2 (\Delta t)^2 + 2ab\, \Delta t\, \Delta z + b^2 (\Delta z)^2.$$

  $\hookrightarrow$ the three terms have orders: $O((\Delta t)^2)$, $O((\Delta t)^{3/2})$, $O(\Delta t)$. As $\Delta t \downarrow 0$, the first two terms vanish.

- In the continuous-time limit, quadratic variation is then summarized by: $(\mathrm{d}t)^2 = 0$, $\mathrm{d}t\mathrm{d}z_t = 0$, $\boxed{(\mathrm{d}z_t)^2 = \mathrm{d}t.}$
- If $\mathrm{d}X_t = a\mathrm{d}t + b\mathrm{d}z_t$, then, to first order in $\mathrm{d}t$,

$$\boxed{(\mathrm{d}X_t)^2 = b^2\, \mathrm{d}t.}$$

$\hookrightarrow$ The second-order Taylor term therefore survives:

$$\frac{1}{2} f_{XX} (\mathrm{d}X_t)^2 = \frac{1}{2} b^2 f_{XX}\, \mathrm{d}t.$$

Uncertainty affects the **drift** of a nonlinear transformation because Brownian variation accumulates at order $\mathrm{d}t$.

Itô's Lemma

- Let the state variable x_t follow the diffusion model

$$\mathrm{d}x_t = \mu_x(x_t)\,\mathrm{d}t + \sigma_x(x_t)\,\mathrm{d}z_t,$$

  and consider a nonlinear transformation $y_t = f(x_t)$.
- From the second-order expansion,

$$\mathrm{d}y_t = f_x\,\mathrm{d}x_t + \frac{1}{2}f_{xx}(\mathrm{d}x_t)^2.$$

  and recall from the previous slide: $(\mathrm{d}x_t)^2 = \sigma_x^2\,\mathrm{d}t$.
- Substituting both terms gives Itô's lemma:

$$\boxed{\mathrm{d}y_t = \left(\mu_x f_x + \frac{1}{2}\sigma_x^2 f_{xx}\right)\mathrm{d}t + \sigma_x f_x\,\mathrm{d}z_t.}$$

↪ A nonlinear transformation therefore changes both parts of the process:

$$\underbrace{\mu_y}_{\text{drift of } y} = \mu_x f_x + \frac{1}{2}\sigma_x^2 f_{xx}, \qquad \underbrace{\sigma_y}_{\text{diffusion of } y} = \sigma_x f_x.$$

  In particular, curvature f_{xx} affects the **expected change** in y.
- If the function also depends explicitly on time, $y_t = f(x_t, t)$, one additional term appears:

$$\mathrm{d}y_t = \left(f_t + \mu_x f_x + \frac{1}{2}\sigma_x^2 f_{xx}\right)\mathrm{d}t + \sigma_x f_x\,\mathrm{d}z_t.$$

Itô Example: Simple Returns vs. Log Returns

- Suppose the risky asset follows $\frac{\mathrm{d}S_t}{S_t} = \mu \mathrm{d}t + \sigma \mathrm{d}z_t$. Thus μ is the drift of the instantaneous simple return.
- We now ask how the log price evolves, $y_t = \log S_t$. For $f(S) = \log S$, we have: $f_S = \frac{1}{S}$,$f_{SS} = -\frac{1}{S^2}$.
- We start by the expansion: $\mathrm{d}\log S_t = \frac{1}{S_t}\mathrm{d}S_t - \frac{1}{2}\frac{1}{S_t^2}(\mathrm{d}S_t)^2$. Since

$$\mathrm{d}S_t = S_t(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}z_t), \qquad (\mathrm{d}S_t)^2 = S_t^2\sigma^2\,\mathrm{d}t,$$

substitution yields (Itô's lemma)

$$\boxed{\mathrm{d}\log S_t = \left(\mu - \frac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}z_t.}$$

- The term $-\sigma^2/2$ comes entirely from the curvature of the logarithm: $f_{SS} < 0$.
  $\hookrightarrow$ Greater volatility therefore lowers expected log growth even though $\mathbb{E}_t\left[\frac{\mathrm{d}S_t}{S_t}\right] = \mu \mathrm{d}t$.

$\hookrightarrow$ Simple and log returns therefore have different drifts:

$$\text{simple-return drift} = \mu, \qquad \text{log-return drift} = \mu - \frac{1}{2}\sigma^2.$$

This is the continuous-time counterpart of the Jensen adjustment.

Dynamic Programming and the HJB

- The principle of dynamic programming is to break a dynamic optimization problem into a sequences. Let X_t denote the state and u_t the control. The value function summarizes the optimal attainable expected utility from the current state onward: $(\rho > 0)$

$$V(X_t, t) = \max_{\{u_s\}_{s \geq t}} \mathbb{E}_t \left[\int_t^{\infty} e^{-\rho(s-t)} U(X_s, u_s) \, \mathrm{d}s \right].$$

- Over a short interval $\mathrm{d}t$, Bellman's principle rewrites the same problem as

$$V(X_t, t) = \max_{u_t} \left\{ U(X_t, u_t) \, \mathrm{d}t + e^{-\rho \mathrm{d}t} \mathbb{E}_t [V(X_{t+\mathrm{d}t}, t + \mathrm{d}t)] \right\}.$$

↪ HJB eq. is obtained by asking what this recursion implies as the interval $\mathrm{d}t$ becomes arbitrarily small.

- Once the current state is known, the agent faces the following trade-off in choosing u_t:

$$\underbrace{U(X_t, u_t)}_{\text{utility today}} \quad \text{and} \quad \underbrace{V(X_{t+\mathrm{d}t}, t + \mathrm{d}t)}_{\text{value of the future state}}.$$

From the Continuous-Time Problem to the HJB

- Let the state follow the diffusion model: $\mathrm{d}X_t = \mu(X_t, u_t, t)\,\mathrm{d}t + \sigma(X_t, u_t, t)\,\mathrm{d}z_t$.
  The control u_t can therefore affect both the expected evolution (μ) and the risk of the state (σ).
- From the previous slide:

$$V(X_t, t) = \max_{u_t} \left\{ U(X_t, u_t)\,\mathrm{d}t + e^{-\rho \mathrm{d}t} \mathbb{E}_t[V(X_{t+\mathrm{d}t}, t + \mathrm{d}t)] \right\}.$$

- For a short interval,

$$e^{-\rho \mathrm{d}t} = 1 - \rho\,\mathrm{d}t + o(\mathrm{d}t).$$

- Once applied Itô's lemma to the continuation value:

$$\mathrm{d}V = \left(V_t + \mu V_X + \frac{1}{2}\sigma^2 V_{XX} \right) \mathrm{d}t + \sigma V_X\,\mathrm{d}z_t.$$

- Since $\mathbb{E}_t[\mathrm{d}z_t] = 0$, the expected change in continuation value is

$$\boxed{\mathbb{E}_t[\mathrm{d}V] = \left(V_t + \mu V_X + \frac{1}{2}\sigma^2 V_{XX} \right) \mathrm{d}t.}$$

↪ We are now ready to substitute the discounting and continuation-value expansions into Bellman's.

From Bellman's Recursion to the HJB

- Hence
$$\mathbb{E}_t[V(X_{t+\mathrm{d}t}, t+\mathrm{d}t)] = V(X_t,t) + \left(V_t + \mu V_X + \frac{1}{2}\sigma^2 V_{XX}\right)\mathrm{d}t + o(\mathrm{d}t).$$
- Substitute these expressions into Bellman's recursion:
$$V = \max_{u_t}\left\{U\,\mathrm{d}t + (1-\rho\mathrm{d}t)\left[V + \left(V_t + \mu V_X + \frac{1}{2}\sigma^2 V_{XX}\right)\mathrm{d}t\right]\right\} + o(\mathrm{d}t).$$
- Keeping only terms of order $\mathrm{d}t$,
$$V = \max_{u_t}\left\{V + \left[U + V_t + \mu V_X + \frac{1}{2}\sigma^2 V_{XX} - \rho V\right]\mathrm{d}t\right\} + o(\mathrm{d}t).$$
- Cancel V, divide by $\mathrm{d}t$, and let $\mathrm{d}t \downarrow 0$:
$$\boxed{0 = \max_{u}\left\{U(X,u) + V_t + \mu(X,u,t)V_X + \frac{1}{2}\sigma^2(X,u,t)V_{XX} - \rho V\right\}.}$$
- If the problem is stationary:
$$\boxed{\rho V = \max_{u}\left\{U + V_t + \mu V_X + \frac{1}{2}\sigma^2 V_{XX}\right\}.}$$
when V is not time dependent then $V_t = 0$.

↪ The HJB is therefore Bellman's optimality condition written locally in continuous time.

How to Read the HJB

$$\rho V = \max_u \left\{ \overbrace{U}^{\text{current utility}} + \overbrace{V_t}^{\text{time effect}} + \overbrace{\mu V_X}^{\text{expected state change}} + \overbrace{\tfrac{1}{2}\sigma^2 V_{XX}}^{\text{effect of local risk}} \right\}$$

How to read the HJB

- ρV is the required instantaneous return on the utility value represented by V.
- μV_X values the expected movement in the state at its marginal continuation value.
- $\frac{1}{2}\sigma^2 V_{XX}$ captures how uncertainty affects continuation value through curvature.
- For a concave value function (i.e., $V_{XX} < 0$), local uncertainty lowers continuation value.

What is missing?

- The HJB determines the shape of V in the **interior**.
- To pin down a particular solution, we also need (economically appropriate) boundary/terminal conditions.
- In finite-horizon, the value function must match the payoff that remains at the terminal date ($V(X,T) = G(X)$).
  $\hookrightarrow$ HJB determines how this terminal value is propagated backward.
- We may also need conditions at the boundaries of the state space. Depending on the economic problem, the boundary may pin down the level of V or impose a condition on its slope.

$\hookrightarrow$ The HJB gives the local optimality condition
VS. boundary/terminal, or later transversality conditions select the economically relevant solution.

The Hamiltonian: the Optimal Control

- The HJB can be written as: $\rho V = V_t + \max_{u \in \mathcal{U}} \left\{ U(X,u) + \mu(X,u,t)V_X + \frac{1}{2}\sigma^2(X,u,t)V_{XX} \right\}$.
- The terms that depend on the control are collected in the **Hamiltonian**:

$$\mathcal{H}(X,t,u) = U(X,u) + \mu(X,u,t)V_X + \frac{1}{2}\sigma^2(X,u,t)V_{XX}.$$

- The optimal control therefore solves: $u^*(X,t) \in \arg\max_{u \in \mathcal{U}} \mathcal{H}(X,t,u)$.
- If the optimum is an interior, the FOC is

$$\left.\frac{\partial \mathcal{H}}{\partial u}\right|_{u=u^*} = 0.$$

  If the control is constrained, the maximization must respect the constraints as well.
- In the Merton problem, the controls will be consumption and the risky portfolio share.
  $\hookrightarrow$ We obtain their optimal policies directly from the Hamiltonian.

The Merton Consumption-Portfolio Problem

- Following Merton (1971), consider an investor who chooses i) how much to consume and ii) how to allocate wealth between a risky asset and a risk-free asset over an infinite horizon.
  $\hookrightarrow$ The investment opportunities are: $\frac{\mathrm{d}A_t}{A_t} = r\mathrm{d}t$, and $\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}z_t$.
- At each date t, the investor chooses: i) the consumption rate, C_t, and ii) the fraction of wealth in the risky asset, ω_t. The remaining fraction $1-\omega_t$ is invested in the risk-free asset.
- The objective is

$$\boxed{V(W_t,t) = \max_{\{C_s,\omega_s\}_{s\geq t}} \mathbb{E}_t\left[\int_t^\infty e^{-\rho(s-t)}U(C_s)\,\mathrm{d}s\right].}$$

- The relevant state variable is financial wealth W_t.
  The controls (C_t,ω_t) determine both the drift and the risk of its evolution.

$\hookrightarrow$ We therefore first derive the wealth dynamics implied by consumption and portfolio choice, and then insert them into the general HJB.

Wealth Dynamics

- With wealth W_t, the investor allocates

$$\omega_t W_t \quad \text{to the risky asset,} \qquad (1-\omega_t)W_t \quad \text{to the risk-free asset.}$$

- Before consumption, the portfolio earns

$$\begin{aligned}(1-\omega_t)W_t\, r\,\mathrm{d}t + \omega_t W_t\,(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}z_t)\\ = [rW_t + \omega_t W_t(\mu - r)]\,\mathrm{d}t + \omega_t W_t \sigma\,\mathrm{d}z_t.\end{aligned}$$

- Consumption is a flow variable: $C_t\,\mathrm{d}t$ units of wealth over the interval $\mathrm{d}t$.
- Hence the investor's wealth evolves according to

$$\mathrm{d}W_t = [rW_t + \omega_t W_t(\mu - r) - C_t]\,\mathrm{d}t + \omega_t W_t \sigma\,\mathrm{d}z_t.$$

- Therefore the wealth drift and variance are

$$\mu_W = rW_t + \omega_t W_t(\mu - r) - C_t, \qquad \sigma_W^2 = \omega_t^2 W_t^2 \sigma^2.$$

↪ Consumption lowers expected wealth growth,
while risky investment affects both i) exp. wealth growth and ii) wealth risk.

The Merton HJB

- From the general continuous-time control problem,

$$\rho V(W,t) = \max_{C,\omega} \left\{ U(C) + V_t + V_W\, \mu_W + \frac{1}{2} V_{WW}\, \sigma_W^2 \right\}.$$

- From the wealth dynamics derived above,

$$\mu_W = rW + \omega W(\mu - r) - C, \qquad \sigma_W^2 = \omega^2 W^2 \sigma^2.$$

- Substituting these expressions into the HJB gives

$$\boxed{\rho V(W,t) = \max_{C \geq 0,\, \omega} \left\{ U(C) + V_t + V_W \left[rW + \omega W(\mu - r) - C \right] + \frac{1}{2} V_{WW} \omega^2 W^2 \sigma^2 \right\}.}$$

- The two controls enter through different margins: i) C enters current utility and wealth drift, ii) ω enters wealth drift and variance.
- Under the time-homogeneous infinite-horizon Merton benchmark, stationarity can be imposed later when solving for the value function.

CRRA Utility and Homogeneity

- Let us consider CRRA utility:
$$U(C) = \frac{C^{1-\gamma}}{1-\gamma}, \qquad \gamma > 0, \quad \gamma \neq 1.$$
- Recall γ is both relative risk aversion ($-\frac{C\,U''(C)}{U'(C)}$) and the inverse of elasticity of intertemporal substitution.

↪ CRRA utility is homogeneous of degree $1-\gamma$: $U(aC) = a^{1-\gamma}U(C)$, with $a > 0$.

- The wealth dynamics are linear in wealth and consumption. Hence, if a policy $\{C_s, \omega_s\}$ is feasible starting from wealth W, then $\{aC_s, \omega_s\}$ is feasible starting from wealth aW.
  ↪ Scaling initial wealth by a scales the value of the entire optimization problem by $a^{1-\gamma}$:
$$V(aW, t) = a^{1-\gamma}V(W, t).$$
- This homogeneity property determines the dependence of the value function on wealth. We write the guess:
$$\boxed{V(W, t) = \frac{1}{1-\gamma}W^{1-\gamma}v(t),}$$
  for some unknown function $v(t)$ collecting the remaining dependence on time.
- The corresponding derivatives are
$$V_t = \frac{W^{1-\gamma}}{1-\gamma}v'(t), \qquad V_W = W^{-\gamma}v(t), \qquad V_{WW} = -\gamma W^{-\gamma-1}v(t).$$

Step-by-Step Solution: the CRRA Guess

- Start from the Merton HJB:

$$\rho V = \max_{C,\omega} \left\{ U(C) + V_t + V_W \big[rW + \omega W(\mu - r) - C \big] + \frac{1}{2} V_{WW} \omega^2 W^2 \sigma^2 \right\}.$$

- Under CRRA, substituting these expressions into the HJB gives

$$\rho \frac{W^{1-\gamma}}{1-\gamma} v(t) = \max_{C,\omega} \left\{ \frac{C^{1-\gamma}}{1-\gamma} + \frac{W^{1-\gamma}}{1-\gamma} v'(t) + W^{-\gamma} v(t) \big[rW + \omega W(\mu - r) - C \big] \right.$$
$$\left. - \frac{1}{2} \gamma W^{1-\gamma} v(t) \omega^2 \sigma^2 \right\}.$$

- Divide both sides by $W^{1-\gamma}/(1-\gamma)$:

$$\rho v(t) = \max_{C,\omega} \left\{ \left(\frac{C}{W} \right)^{1-\gamma} + v'(t) + (1-\gamma) v(t) \left[r + \omega(\mu - r) - \frac{C}{W} \right] - \frac{1}{2} \gamma (1-\gamma) v(t) \omega^2 \sigma^2 \right\}.$$

↪ Homogeneity removes the level of wealth: the choice of consumption now enters only through the ratio C/W.

Step-by-Step HJB Solution: Optimal Controls

- From the normalized HJB, the terms involving consumption are

$$\left(\frac{C}{W}\right)^{1-\gamma} - (1-\gamma)v(t)\frac{C}{W}.$$

The first-order condition for C is therefore

$$0 = \frac{\partial}{\partial C}\left[\left(\frac{C}{W}\right)^{1-\gamma} - (1-\gamma)v(t)\frac{C}{W}\right] = \frac{1-\gamma}{W}\left[\left(\frac{C}{W}\right)^{-\gamma} - v(t)\right] \iff \frac{C^*}{W} = v(t)^{-1/\gamma}.$$

↪ Optimal consumption is therefore proportional to wealth, with a consumption–wealth ratio governed by $v(t)$.

- The terms involving the portfolio share are

$$(1-\gamma)v(t)\left[\omega(\mu - r) - \frac{1}{2}\gamma\omega^2\sigma^2\right].$$

The first-order condition for ω gives

$$(\mu - r) - \gamma\omega\sigma^2 = 0 \iff \omega^* = \frac{\mu - r}{\gamma\sigma^2}$$

↪ The risky share is already determined by investment opportunities and risk aversion, while the consumption rule still depends on the unknown function $v(t)$.

▷ We now substitute both optimal controls back into the HJB to determine the evolution of $v(t)$.

Step-by-Step HJB Solution: Back to the HJB

- At the optimal consumption choice,

$$\left(\frac{C^*}{W}\right)^{1-\gamma} - (1-\gamma)v(t)\frac{C^*}{W} = v(t)^{\,1-1/\gamma} - (1-\gamma)v(t)^{\,1-1/\gamma} = \gamma v(t)^{\,1-1/\gamma}.$$

- At the optimal portfolio choice,

$$\omega^*(\mu - r) - \frac{1}{2}\gamma(\omega^*)^2\sigma^2 = \frac{(\mu-r)^2}{2\gamma\sigma^2}.$$

- Substituting both optimal controls into the normalized HJB gives

$$\rho v(t) = v'(t) + (1-\gamma)\left[r + \frac{(\mu-r)^2}{2\gamma\sigma^2}\right] v(t) + \gamma v(t)^{\,1-1/\gamma}.$$

- Equivalently,

$$\boxed{v'(t) = \left[\rho - (1-\gamma)r - (1-\gamma)\frac{(\mu-r)^2}{2\gamma\sigma^2}\right] v(t) - \gamma v(t)^{\,1-1/\gamma}.}$$

↪ The optimization over (C, ω) is done (Hamiltonian).
What remains is a (nonlinear) differential equation for the time component $v(t)$ of the value function.

Step-by-Step HJB Solution: Linearizing the ODE

- Write:
$$v'(t) = a\,v(t) - \gamma v(t)^{\,1-1/\gamma},$$
where $a := \rho - (1-\gamma)r - (1-\gamma)(\mu - r)^2/(2\gamma\sigma^2)$.
- The equation is nonlinear because of the term $v(t)^{\,1-1/\gamma}$. To simplify, let us rewrite it:
$$y(t) := v(t)^{1/\gamma} \qquad \Longleftrightarrow \qquad v(t) = y(t)^{\gamma}.$$
- Differentiating,
$$v'(t) = \gamma y(t)^{\gamma-1} y'(t).$$
- Substituting $v(t) = y(t)^{\gamma}$ and $v'(t) = \gamma y(t)^{\gamma-1} y'(t)$ into the ODE gives
$$\gamma y^{\gamma-1} y' = a y^{\gamma} - \gamma y^{\gamma-1}.$$
- Dividing by $\gamma y^{\gamma-1}$ yields the linear equation
$$\boxed{y'(t) = \frac{a}{\gamma} y(t) - 1.}$$

↪ The transformation $y(t) = v(t)^{1/\gamma}$ converts the nonlinear HJB equation into a first-order linear ODE.

▷ Under stationarity, the time component is constant, so the next step is to impose $y'(t) = 0$ and recover a constant consumption–wealth ratio.

Closed-Form and Stationary Solution

- For $a \neq 0$, the linear ODE, $y'(t) = \frac{a}{\gamma} y(t) - 1$, has the general solution: $y(t) = \frac{\gamma}{a} + \left(y(0) - \frac{\gamma}{a}\right) e^{\frac{a}{\gamma} t}$.
  Since $y(t) = v(t)^{1/\gamma}$,

$$\boxed{v(t) = \left[\frac{\gamma}{a} + \left(v(0)^{1/\gamma} - \frac{\gamma}{a}\right) e^{\frac{a}{\gamma} t}\right]^{\gamma}.}$$

- In the stationary infinite-horizon Merton problem, the value function does not depend on time.
  Hence we select the constant solution:

$$y(t) \equiv \bar{y} \qquad \Longleftrightarrow \qquad y'(t) = 0.$$

- From the ODE, $0 = \frac{a}{\gamma} \bar{y} - 1 \implies \bar{y} = \frac{\gamma}{a}$,
  which is equivalently obtained from the general solution by setting: $y(0) = \gamma/a$.
- Therefore

$$\boxed{\bar{v} = \left(\frac{\gamma}{a}\right)^{\gamma},} \qquad \boxed{\frac{C_t^*}{W_t} = \bar{v}^{-1/\gamma} = \frac{a}{\gamma} \equiv \kappa.}$$

↪ The general ODE allows a time-varying value-function coefficient;
stationarity selects its constant solution and therefore a constant consumption–wealth ratio.

The Stationary Merton Solution

- Recall: $a := \rho - (1-\gamma)r - (1-\gamma)\frac{(\mu-r)^2}{2\gamma\sigma^2}$.
- The stationary value-function coefficient is: $\bar{v} = \left(\frac{\gamma}{a}\right)^{\gamma}$.
- The two optimal policies are therefore

$$\boxed{\frac{C_t^*}{W_t} = \kappa = \frac{a}{\gamma},} \qquad \boxed{\omega_t^* = \frac{\mu - r}{\gamma\sigma^2}.}$$

- Equivalently, the optimal consumption–wealth ratio is

$$\kappa = \frac{1}{\gamma}\left[\rho - (1-\gamma)r - (1-\gamma)\frac{(\mu-r)^2}{2\gamma\sigma^2}\right].$$

- Hence both optimal policies are **scale free**:

$$C_t^* = \kappa W_t, \qquad \omega_t^* = \text{constant}.$$

  Wealth changes the level of consumption and risky investment, but not their proportions relative to wealth.
- A stationary solution with positive consumption requires

$$\kappa > 0 \quad \Longleftrightarrow \quad a > 0,$$

  or equivalently

$$\rho > (1-\gamma)r + (1-\gamma)\frac{(\mu-r)^2}{2\gamma\sigma^2}.$$

↪ The Merton problem therefore delivers two simple rules:
i) a constant fraction of wealth is consumed, and ii) a constant fraction is invested in the risky asset.

Consumption-Wealth Ratio: Comparative Statics

- It is useful to summarize the investment certainty-equivalent opportunity set by $g^* := r + \frac{(\mu - r)^2}{2\gamma\sigma^2}$.
  $\hookrightarrow$ safe return + the utility value of access to the risky investment opportunity.
- The stationary consumption–wealth ratio can then be written as:

$$\kappa = \frac{C_t^*}{W_t} = \frac{\rho - (1-\gamma)g^*}{\gamma} = \frac{\rho}{\gamma} + \frac{\gamma - 1}{\gamma} g^*.$$

  - Greater impatience raises current consumption: $\frac{\partial \kappa}{\partial \rho} = \frac{1}{\gamma} > 0$.
  - Better investment opportunities interacts with the EIS, $1/\gamma$: $\frac{\partial \kappa}{\partial g^*} = \frac{\gamma - 1}{\gamma}$.

$$\begin{cases} \gamma > 1: & g^* \uparrow \Longrightarrow \kappa \uparrow, \\ 0 < \gamma < 1: & g^* \uparrow \Longrightarrow \kappa \downarrow. \end{cases}$$

    $\hookrightarrow$ Better opportunities generate income vs. substit. effects; which dominates depends on the EIS.
- A stationary solution with positive consumption requires

$$\boxed{\kappa > 0 \quad \Longleftrightarrow \quad \rho > (1-\gamma)g^*.}$$

For $0 < \gamma < 1$, sufficiently attractive investment opportunities can otherwise make the infinite-horizon utility problem unbounded. For $\gamma > 1$ and conventional positive g^*, the condition is automatically satisfied.

$\hookrightarrow$ Portfolio choice responds directly to the reward-to-risk trade-off; consumption also reflects the investor's willingness to substitute consumption across time.

Value Function, Admissibility, Transversality

- From the stationary solution, $\bar{v} = \left(\frac{\gamma}{a}\right)^{\gamma} = \kappa^{-\gamma}, \kappa = \frac{a}{\gamma}$. Hence the value function is

$$\boxed{V(W) = \frac{\kappa^{-\gamma}}{1-\gamma} W^{1-\gamma}.}$$

- Under the optimal policies, $C_t^* = \kappa W_t, \omega^* = \frac{\mu - r}{\gamma \sigma^2}$, wealth follows a geometric diffusion:

$$\frac{\mathrm{d}W_t}{W_t} = \left[r + \omega^*(\mu - r) - \kappa\right] \mathrm{d}t + \omega^* \sigma \, \mathrm{d}z_t.$$

- Solving the HJB and the first-order conditions is not sufficient for an infinite-horizon problem.
  ↪ We must also verify that the discounted continuation value vanishes:

$$\boxed{\lim_{T \to \infty} \mathbb{E}\left[e^{-\rho T} V(W_T^*)\right] = 0.}$$

- For the CRRA candidate and the optimal wealth process, $\mathbb{E}\left[e^{-\rho T} V(W_T^*)\right] = V(W_0) e^{-\kappa T}$. Therefore,

$$\kappa > 0 \quad \Longrightarrow \quad \mathbb{E}\left[e^{-\rho T} V(W_T^*)\right] \longrightarrow 0.$$

↪ The same condition $\kappa > 0$ guarantees: i) a positive stationary consumption rate, and ii) the discounted value of remaining wealth vanishes asymptotically.

▷ We have therefore obtained and verified the stationary Merton solution.

In-Class Exercise: Solve and Stress-Test

- Consider the following annualized investment opportunities and preferences:

	r	μ	σ	ρ	γ
Baseline	2%	8%	20%	4%	3

- Recall the stationary Merton policies:

$$\omega^* = \frac{\mu - r}{\gamma \sigma^2}, \qquad \kappa = \frac{C^*}{W} = \frac{\rho - (1-\gamma) g^*}{\gamma}.$$

- Take 10–15 minutes to work through the following:
  1. Compute the optimal risky wealth share ω^*.
  2. Compute the certainty-equivalent investment opportunity g^* and the optimal consumption–wealth ratio κ.
  3. Verify that the stationary solution is well defined: $\kappa > 0 \iff \rho > (1-\gamma) g^*$.
  4. Increase risk aversion from $\gamma = 3$ to $\gamma = 6$, keeping (r, μ, σ, ρ) unchanged, and recompute ω^*, g^*, and κ.
  5. Compare the two cases economically: which policy reacts more strongly to the increase in γ, and why?

↪ Keep all rates in decimal form during the calculations and interpret ω^* and κ as fractions of wealth.

Exercise Solution (1/2): Baseline

- With
$$r = 0.02, \qquad \mu = 0.08, \qquad \sigma = 0.20, \qquad \rho = 0.04, \qquad \gamma = 3,$$
we have: $\mu - r = 0.06$ and $\sigma^2 = 0.04$. The optimal risky share is therefore:
$$\boxed{\omega^* = \frac{0.06}{3(0.04)} = 0.50.}$$
Thus the investor allocates 50% of wealth to the risky asset.
- The certainty-equivalent investment opportunity is
$$g^* = 0.02 + \frac{0.06^2}{2(3)(0.04)} = 0.035.$$
- Hence the optimal consumption–wealth ratio is
$$\kappa = \frac{0.04 - (1-3)(0.035)}{3} = 0.037.$$
The investor therefore consumes approximately 3.67% of wealth per year.
- Finally,
$$\rho - (1-\gamma)g^* = 0.04 + 2(0.035) = 0.11 > 0,$$
so the stationary solution satisfies the well-posedness condition.

Exercise Solution (2/2): Risk Aversion

- Now double the risk aversion to $\gamma = 6$, holding (r, μ, σ, ρ) fixed.
- The optimal risky share becomes $\omega^* = \frac{0.06}{6(0.04)} = 0.25$, while $g^* = 0.02 + \frac{0.06^2}{2(6)(0.04)} = 0.0275$.
- The consumption–wealth ratio is now: $\kappa = \frac{0.04-(1-6)(0.0275)}{6} = 0.0296$.
- Comparing the two cases:

	$\gamma = 3$	$\gamma = 6$
ω^*	50%	25%
g^*	3.50%	2.75%
κ	3.67%	2.96%

- The economic effects are different:
  - Risky exposure halves because
$$\omega^* \propto \frac{1}{\gamma}.$$
  - Consumption also falls, but less sharply. Changing γ affects consumption through several channels: intertemporal substitution, the direct $1/\gamma$ scaling of κ, and the value of the risky investment opportunity through g^*.
  - Well-posedness continues to hold:
$$\rho - (1-\gamma)g^* = 0.1775 > 0.$$

↪ Portfolio choice reacts directly to risk aversion; consumption reflects both risk preferences and intertemporal substitution.

Log Utility: An Internal Consistency Check

- As a consistency check on the CRRA solution, consider $U(C) = \log C$, and guess a stationary value function of the form $V(W) = A \log W + b$, where we have: $V_W = \frac{A}{W}$, $V_{WW} = -\frac{A}{W^2}$.
- Recall that the stationary Merton HJB is

$$\rho V = \max_{C,\omega} \left\{ \log C + V_W \left[rW + \omega W(\mu - r) - C \right] + \frac{1}{2} V_{WW} \omega^2 W^2 \sigma^2 \right\}.$$

- Write consumption as a fraction of wealth, $C = cW$, meaning that: $\log C = \log c + \log W$, so we write:

$$\rho\big(A \log W + b\big) = \max_{c,\omega} \left\{ \log c + \log W + A\big[r + \omega(\mu - r) - c\big] - \frac{1}{2} A \omega^2 \sigma^2 \right\}.$$

- The only terms involving $\log W$ are $\rho A \log W$ and $\log W$. Since the equality must hold for every $W > 0$,

$$A = \frac{1}{\rho}, \qquad V(W) = \frac{1}{\rho} \log W + b.$$

↪ We can now recover the optimal consumption and portfolio rules and then compare with the CRRA limit $\gamma \to 1$.

Log Utility: Optimal Policies and the CRRA Limit

- The first-order condition for consumption is $\frac{1}{C} - V_W = 0 \quad \Longrightarrow \quad \frac{1}{C} = \frac{1}{\rho W}$. Meaning that: $C^* = \rho W$.
- The portfolio first-order condition is $V_W W(\mu - r) + V_{WW}\omega W^2\sigma^2 = 0$.
  $\hookrightarrow$ Substituting the derivatives, we reach: $\omega^* = \frac{\mu - r}{\sigma^2}$.
- Substituting the optimal controls back into the HJB gives $\rho b = \log \rho + \frac{1}{\rho}\left[r - \rho + \frac{(\mu - r)^2}{2\sigma^2}\right]$, and therefore

$$\boxed{b = \frac{\log \rho}{\rho} + (r + \frac{(\mu - r)^2}{2\sigma^2} - \rho)\rho^{-2}.}$$

- Finally, the CRRA policies converge to the same expressions as $\gamma \to 1$:

$$\omega^* = \frac{\mu - r}{\gamma\sigma^2} \longrightarrow \frac{\mu - r}{\sigma^2},$$

$$\kappa = \frac{\rho - (1 - \gamma)g^*}{\gamma} \longrightarrow \rho.$$

$\hookrightarrow$ Log utility is therefore consistent: the CRRA solution converges smoothly to

$$\frac{C^*}{W} = \rho, \qquad \omega^* = \frac{\mu - r}{\sigma^2}.$$

From Marginal Utility to a State-Price Density

- In Session 2, marginal utility generated the discrete-time SDF: $M_{t+1} = \beta u'(C_{t+1})/u'(C_t)$.
  In continuous time we have the **state-price density**: $\xi_t = e^{-\rho t} u'(C_t)$.
- Write: $\frac{\mathrm{d}\xi_t}{\xi_t} = \mu_{\xi,t}\,\mathrm{d}t - \lambda_t\,\mathrm{d}z_t$, with λ_t the exposure of marginal utility to the Brownian shock.
- The risk-free asset must also satisfy the SDF pricing relation.
  $\hookrightarrow$ As $\frac{\mathrm{d}A_t}{A_t} = r\,\mathrm{d}t$, the product $\xi_t A_t$ must have zero conditional drift: $\mathbb{E}_t[\mathrm{d}(\xi_t A_t)] = 0$.
- Because the risk-free asset has no Brownian component,

$$\frac{\mathrm{d}(\xi_t A_t)}{\xi_t A_t} = \frac{\mathrm{d}\xi_t}{\xi_t} + \frac{\mathrm{d}A_t}{A_t} = (\mu_{\xi,t} + r)\mathrm{d}t - \lambda_t\,\mathrm{d}z_t. \quad \Longrightarrow\ \mu_{\xi,t} = -r$$

  Hence: $\frac{\mathrm{d}\xi_t}{\xi_t} = -r\,\mathrm{d}t - \lambda_t\,\mathrm{d}z_t$.
- The same pricing logic applies to a risky traded asset. For simplicity, suppose the asset pays no dividends. Then its price expressed in state-price units, $\xi_t S_t$, must have zero conditional drift: $\mathbb{E}_t[\mathrm{d}(\xi_t S_t)] = 0$.
- To compute the change in $\xi_t S_t$, we use Itô's product rule:

$$\mathrm{d}(\xi_t S_t) = \xi_t\,\mathrm{d}S_t + S_t\,\mathrm{d}\xi_t + \mathrm{d}\xi_t\,\mathrm{d}S_t.$$

  $\hookrightarrow$ Both ξ_t and S_t contain Brownian shocks (the $\mathrm{d}z_t$ terms), so their covariation is: $\frac{\mathrm{d}\xi_t}{\xi_t}\frac{\mathrm{d}S_t}{S_t} = -\lambda_t\sigma(\mathrm{d}z_t)^2 = -\lambda_t\sigma\,\mathrm{d}t$.

$\hookrightarrow$ Unlike in ordinary calculus, the product of the two shocks survives at order $\mathrm{d}t$. It changes the drift of the state-price-adjusted asset price and is therefore the key term linking risk exposure σ to expected excess returns.

The Dynamic Price of Risk

- Recall the dynamics of the risky asset and the state-price density:

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}z_t, \qquad \frac{\mathrm{d}\xi_t}{\xi_t} = -r\,\mathrm{d}t - \lambda_t\,\mathrm{d}z_t.$$

- Using Itô's product rule,

$$\frac{\mathrm{d}(\xi_t S_t)}{\xi_t S_t} = \frac{\mathrm{d}\xi_t}{\xi_t} + \frac{\mathrm{d}S_t}{S_t} + \frac{\mathrm{d}\xi_t}{\xi_t}\frac{\mathrm{d}S_t}{S_t}.$$

- Since $\frac{\mathrm{d}\xi_t}{\xi_t}\frac{\mathrm{d}S_t}{S_t} = -\lambda_t\sigma\,\mathrm{d}t$, we obtain: $\frac{\mathrm{d}(\xi_t S_t)}{\xi_t S_t} = \big(\mu - r - \lambda_t\sigma\big)\mathrm{d}t + \big(\sigma - \lambda_t\big)\mathrm{d}z_t$.
- The pricing condition requires the state-price-adjusted asset price to have zero conditional drift: $\mathbb{E}_t[\mathrm{d}(\xi_t S_t)] = 0$.

↪ The expected excess return therefore equals

$$\mu - r = \underbrace{\lambda_t}_{\substack{\text{price per unit}\\ \text{of Brownian risk}}} \times \underbrace{\sigma}_{\text{quantity of Brownian risk}}$$

- Using the pricing relation and the Merton portfolio rule, $\omega_t^* = \frac{\mu - r}{\gamma\sigma^2}$, we reach: $\omega_t^* = \frac{\lambda_t}{\gamma\sigma}$.
  Equivalently, the investor's optimal Brownian exposure is

$$\boxed{\omega_t^*\sigma = \frac{\lambda_t}{\gamma}.}$$

↪ The same λ_t links **pricing** and **choice**: markets reward Brownian risk at rate λ_t, while risk tolerance $1/\gamma$ determines how much of that risk the investor chooses to bear.

Intertemporal Hedging Demand

- So far, (r, μ, σ) were constant. Suppose instead that an additional state variable X_t changes future expected returns or volatility:

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + s_X(X_t)\,\mathrm{d}z_t^X, \qquad \mathrm{d}z_t\,\mathrm{d}z_t^X = \eta\,\mathrm{d}t.$$

  Continuation value now depends on both wealth and the investment-opportunity state: $V = V(W, X)$.
- As wealth and the state can move together, their quadratic covariation is $\mathrm{d}W_t\,\mathrm{d}X_t = \omega W \sigma s_X \eta\,\mathrm{d}t$.

↪ Itô's lemma therefore adds the following cross term to the HJB:

$$\boxed{V_{WX}\,\omega W \sigma s_X \eta}$$

↪ how a portfolio shock affects wealth precisely when future investment opportunities change.

- The portfolio first-order condition becomes $V_W W(\mu - r) + V_{WW}\omega W^2\sigma^2 + V_{WX} W \sigma s_X \eta = 0$, so

$$\boxed{\omega^* = -\frac{V_W}{W V_{WW}}\frac{\mu - r}{\sigma^2} - \frac{V_{WX}}{W V_{WW}}\frac{s_X \eta}{\sigma}.}$$

- The optimal risky position now has two components:

$$\underbrace{-\frac{V_W}{W V_{WW}}\frac{\mu - r}{\sigma^2}}_{\text{"myopic" demand}} + \underbrace{-\frac{V_{WX}}{W V_{WW}}\frac{s_X \eta}{\sigma}}_{\text{intertemporal hedging demand}}.$$

↪ The hedging term appears because the risky asset can transfer wealth across states with different **future investment opportunities**. If $V_{WX} = 0$ or $\eta = 0$, it disappears and the constant-opportunity Merton rule returns.

What Have We Learned?

- Continuous time describes local asset-price dynamics through drift and diffusion: $\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}z_t$, while quadratic variation, $(\mathrm{d}z_t)^2 = \mathrm{d}t$, makes curvature matter locally through Itô's lemma.
- Dynamic programming converts the intertemporal problem into the HJB. Under CRRA utility and constant investment opportunities, the Merton problem is solved as:

$$\boxed{\omega^* = \frac{\mu - r}{\gamma\sigma^2},} \qquad \boxed{\frac{C^*}{W} = \frac{\rho - (1-\gamma)\left(r + \frac{(\mu-r)^2}{2\gamma\sigma^2}\right)}{\gamma}.}$$

↪ An appealing simple structure: $\underbrace{W}_{\text{only state}} \implies \underbrace{\text{constant consumption–wealth ratio}}_{\kappa}$ and $\underbrace{\text{risky demand}}_{\omega^*}$.

- Dynamic pricing connects the same investment opportunities to asset prices: $\mu - r = \lambda_t \sigma$.
  The market price of risk λ_t determines compensation for Brownian exposure, while risk tolerance determines how much exposure the investor chooses.
- Once investment opportunities become stochastic, $V = V(W, X)$, and risky demand becomes

$$\boxed{\text{"myopic" demand} + \text{intertemporal hedging demand.}}$$

↪ The benchmark is useful because every extension has a visible entry point: preferences, wealth dynamics, additional state variables, or market constraints.

From Investors to Firms

- In this session, we treated (r, μ, σ) as investment opportunities available to the investor and asked how optimal consumption and portfolio choices respond to them.
- The investor side gave us two complementary objects:

optimal portfolio choice and state-price density / SDF.

- Session 4 reverses the perspective: we ask how firms' investment decisions generate economically relevant returns.
- Pricing logic remains the same but new state and choice variables will include

physical capital, investment, adjustment costs, marginal q.

↪ Session 3 asked how investors choose given investment opportunities.

▷ Session 4 asks how firms' production and investment decisions help determine those returns: **Production-Based Asset Pricing.**

Thanks for your attention! See you next time!

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